

FINARTS HW 3 — Market Model Regression

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1 Overview

Model:

$$R_i = \beta_0 + \beta_1 \cdot R_m + u$$

Data files: - `Company_Returns.xlsx` — Daily returns of all active PSE-listed companies (Jan 1, 2023 – Jan 23, 2026), long format - `PSEi_Return.xlsx` — Daily returns of the PSE index (Jan 1, 2023 – Jan 29, 2025) - `Company Name.xlsx` — Company details (name, RIC, sector)

Selected stocks (5 sectors):

Stock	RIC	Sector	Company
BPI	BPI.PS	Banks	Bank of the Philippine Islands
Century Pacific	CNPF.PS	Food Producers	Century Pacific Food
Meralco	MER.PS	Electricity	Manila Electric (Meralco)
ABS-CBN	ABS.PS	Media	ABS-CBN
PLDT	TEL.PS	Telecommunications	PLDT

2 Data Loading

```
library(dplyr)
library(tidyr)
library(readxl)
library(broom)

set.seed(42L)

company_returns <- read_excel("data/Company_Returns.xlsx")
psei_return      <- read_excel("data/PSEi_Return.xlsx")
company_info     <- read_excel("data/Company Name.xlsx")

cat("DATA LOADED\n")
```

DATA LOADED

```
cat("Company_Returns:", nrow(company_returns), "rows\n")
```

Company_Returns: 204946 rows

```
cat("PSEi_Return:", nrow(psei_return), "rows\n")
```

PSEi_Return: 730 rows

```
cat("Company Info:", nrow(company_info), "rows\n\n")
```

Company Info: 280 rows

```
selected_rics <- c("BPI.PS", "CNP.F.PS", "MER.PS", "ABS.PS", "TEL.PS")

stock_meta <- company_info |>
  filter(RIC %in% selected_rics) |>
  select(RIC, Name = Name, Sector, Full_Name = `Full Name`)

cat("SELECTED STOCKS\n")
```

SELECTED STOCKS

```
print(stock_meta, row.names = FALSE)
```

```
# A tibble: 5 x 4
  RIC      Name      Sector      Full_Name
  <chr>   <chr>      <chr>      <chr>
1 BPI.PS BANK OF THE PHILP.ISLE. Banks      Bank of ~
2 MER.PS MANILA ELECTRIC      Electricity Manila E~
3 TEL.PS PLDT      Telecommunications Service Providers PLDT
4 CNPF.PS CENTURY PACIFIC FOOD Food Producers Century ~
5 ABS.PS ABS-CBN      Media      ABS-CBN
```

```
cat("\n")
```

3 PART 1: Regression Analysis

Instruction: Select 5 stocks from 5 different sectors. Run a regression of the returns of each stock against the market returns.

$$R_i = \beta_0 + \beta_1 \cdot R_m + u$$

3.1 Question 1 (2 pts)

Create a dataframe for your regressions. It must contain the 5 stock returns, PSEi return, and date.

```
psei <- psei_return |>
  mutate(Date = as.Date(Date)) |>
  rename(PSEi_Return = PSEi_Return) |>
  distinct(Date, .keep_all = TRUE)

stock_returns <- company_returns |>
  filter(RIC %in% selected_rics) |>
  rename(Return = Total_Return, Date = Date) |>
  mutate(Date = as.Date(Date))

reg_data_long <- stock_returns |>
  inner_join(psei, by = "Date")

cat("REGRESSION DATA (long format)\n")
```

REGRESSION DATA (long format)

```
cat("Rows:", nrow(reg_data_long), "\n")
```

Rows: 3635

```
cat("Date range:", format(min(reg_data_long$Date), "%Y-%m-%d"),
    "to", format(max(reg_data_long$Date), "%Y-%m-%d"), "\n\n")
```

Date range: 2023-01-03 to 2025-12-29

```
reg_data_wide <- reg_data_long |>
  select(Date, RIC, Return, PSEi_Return) |>
  pivot_wider(
    id_cols      = c(Date, PSEi_Return),
    names_from   = RIC,
    values_from  = Return,
    names_prefix = ""
  ) |>
  mutate(across(all_of(selected_rics), ~ as.numeric(.)))

stopifnot(all(sapply(reg_data_wide[selected_rics], is.numeric)))

cat("REGRESSION DATAFRAME (wide format)\n")
```

REGRESSION DATAFRAME (wide format)

```
cat("Dimensions:", nrow(reg_data_wide), "rows x", ncol(reg_data_wide), "cols\n")
```

Dimensions: 727 rows x 7 cols

```
cat("Columns:", colnames(reg_data_wide), "\n")
```

Columns: Date PSEi_Return BPI.PS MER.PS TEL.PS CNPF.PS ABS.PS

```
cat("First 5 rows shown below.\n\n")
```

First 5 rows shown below.

3.2 Question 2 (2 pts each, 10 pts total)

Run the regressions and report your estimated parameters. Follow this format when reporting:

```

STOCK 1
RIC: XX
Company Name: XX
Sector: XX
Estimated parameters:
B0: XX (indicate significance with asterisks)
B1: XX (indicate significance with asterisks)

```

```

stock_names <- list(
  "BPI.PS" = "Bank of the Philippine Islands",
  "CNP.F.PS" = "Century Pacific Food",
  "MER.PS" = "Manila Electric (Meralco)",
  "ABS.PS" = "ABS-CBN",
  "TEL.PS" = "PLDT"
)

stock_sectors <- list(
  "BPI.PS" = "Banks",
  "CNP.F.PS" = "Food Producers",
  "MER.PS" = "Electricity",
  "ABS.PS" = "Media",
  "TEL.PS" = "Telecommunications Service Providers"
)

sig_stars <- function(p_value) {
  if (is.na(p_value)) return("")
  if (p_value < 0.01) return("***")
  if (p_value < 0.05) return("**")
  if (p_value < 0.10) return("*")
  return("")
}

regression_results <- list()

cat("REGRESSION RESULTS\n")

```

REGRESSION RESULTS

```
cat("Model: R_i = B0 + B1 * R_m + u\n\n")
```

Model: $R_i = B_0 + B_1 * R_m + u$

```

for (ric in selected_rics) {
  stock_data <- reg_data_long |> filter(RIC == ric)
  model <- lm(Return ~ PSEi_Return, data = stock_data)
  regression_results[[ric]] <- model

  coef_summary <- summary(model)$coefficients
  b0_est <- coef_summary[1, 1]
  b1_est <- coef_summary[2, 1]
  b0_pval <- coef_summary[1, 4]
  b1_pval <- coef_summary[2, 4]
  b0_sig <- sig_stars(b0_pval)
  b1_sig <- sig_stars(b1_pval)
  r_squared <- summary(model)$r.squared
  n_obs <- nobs(model)

  cat("STOCK:", which(selected_rics == ric), "\n")
  cat("RIC:", ric, "\n")
  cat("Company Name:", stock_names[[ric]], "\n")
  cat("Sector:", stock_sectors[[ric]], "\n")
  cat("Estimated parameters:\n")
  cat("B0:", sprintf("%.6f%s", b0_est, b0_sig), "\n")
  cat("B1:", sprintf("%.6f%s", b1_est, b1_sig), "\n")
  cat("R-squared:", sprintf("%.4f", r_squared), "\n")
  cat("Observations:", n_obs, "\n\n")
}

```

```

STOCK: 1
RIC: BPI.PS
Company Name: Bank of the Philippine Islands
Sector: Banks
Estimated parameters:
B0: 0.062712
B1: 0.014535
R-squared: 0.0002
Observations: 727

```

```

STOCK: 2
RIC: CNPF.PS
Company Name: Century Pacific Food
Sector: Food Producers
Estimated parameters:
B0: 0.087240

```

B1: -0.029359
R-squared: 0.0005
Observations: 727

STOCK: 3
RIC: MER.PS
Company Name: Manila Electric (Meralco)
Sector: Electricity
Estimated parameters:
B0: 0.120955*
B1: 0.033800
R-squared: 0.0010
Observations: 727

STOCK: 4
RIC: ABS.PS
Company Name: ABS-CBN
Sector: Media
Estimated parameters:
B0: -0.025594
B1: 0.023995
R-squared: 0.0001
Observations: 727

STOCK: 5
RIC: TEL.PS
Company Name: PLDT
Sector: Telecommunications Service Providers
Estimated parameters:
B0: 0.032580
B1: 0.065881*
R-squared: 0.0041
Observations: 727

3.2.1 Detailed Output Per Stock

```
cat("DETAILED MODEL SUMMARIES\n\n")
```

DETAILED MODEL SUMMARIES


```

for (ric in selected_rics) {
  cat("\n", ric, "\n")
  print(summary(regression_results[[ric]]))
  cat("\n")
}

```

BPI.PS

Call:

```
lm(formula = Return ~ PSEi_Return, data = stock_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.7811	-1.1337	-0.0627	1.0006	7.1709

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.06271	0.06897	0.909	0.364
PSEi_Return	0.01453	0.04219	0.344	0.731

Residual standard error: 1.858 on 725 degrees of freedom

Multiple R-squared: 0.0001637, Adjusted R-squared: -0.001215

F-statistic: 0.1187 on 1 and 725 DF, p-value: 0.7306

CNPF.PS

Call:

```
lm(formula = Return ~ PSEi_Return, data = stock_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.264	-1.259	-0.082	1.116	10.381

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.08724	0.07808	1.117	0.264
PSEi_Return	-0.02936	0.04776	-0.615	0.539

Residual standard error: 2.103 on 725 degrees of freedom

Multiple R-squared: 0.0005209, Adjusted R-squared: -0.0008577
 F-statistic: 0.3778 on 1 and 725 DF, p-value: 0.539

MER.PS

Call:

lm(formula = Return ~ PSEi_Return, data = stock_data)

Residuals:

Min	1Q	Median	3Q	Max
-7.2427	-1.0341	-0.1177	0.8704	11.1301

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.12095	0.06554	1.846	0.0654 .
PSEi_Return	0.03380	0.04009	0.843	0.3995

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.765 on 725 degrees of freedom

Multiple R-squared: 0.0009795, Adjusted R-squared: -0.0003985

F-statistic: 0.7108 on 1 and 725 DF, p-value: 0.3995

ABS.PS

Call:

lm(formula = Return ~ PSEi_Return, data = stock_data)

Residuals:

Min	1Q	Median	3Q	Max
-16.4288	-1.4780	-0.1744	0.9323	27.5527

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.02559	0.12647	-0.202	0.840
PSEi_Return	0.02399	0.07736	0.310	0.757

Residual standard error: 3.407 on 725 degrees of freedom

Multiple R-squared: 0.0001327, Adjusted R-squared: -0.001246

F-statistic: 0.09619 on 1 and 725 DF, p-value: 0.7565

TEL.PS

Call:

```
lm(formula = Return ~ PSEi_Return, data = stock_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0471	-0.9067	-0.0326	0.8437	7.4760

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.03258	0.06197	0.526	0.5993
PSEi_Return	0.06588	0.03791	1.738	0.0827 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.669 on 725 degrees of freedom

Multiple R-squared: 0.004148, Adjusted R-squared: 0.002775

F-statistic: 3.02 on 1 and 725 DF, p-value: 0.08267

3.3 Question 3 (5 pts)

Discuss your findings. Discuss the results for each stock, and compare the estimated parameters for each of the stocks.

Per-stock analysis:

Century Pacific Food (CNPF.PS) is the only stock with a negative beta value. A 1% increase in the PSEi leads to a 0.029% decrease in Century Pacific's return. This suggests a defensive characteristic, though the relationship is not statistically significant.

Cross-stock comparison:

Every stock's p-value is greater than 0.05, meaning the results are statistically insignificant. This indicates there is no correlation between our chosen stocks' returns and the PSEi returns from 2023 to 2025. Similarly, the R-squared results for each stock are close to zero, implying the PSEi returns only explain a tiny portion of the variance of price movements for each stock.

4 PART 2: Fitted Values and Residuals

4.1 Question 1 (5 pts)

Using the regression dataframe that you used in Part 1, create columns where you compute the fitted values of the 5 stocks that you analyzed.

$$\hat{R}_{i,t} = \hat{\beta}_0 + \hat{\beta}_1 \cdot R_{m,t}$$

```
reg_data_wide <- reg_data_wide |>
  mutate(
    BPI_Fitted = fitted(regression_results[["BPI.PS"]]),
    CNPF_Fitted = fitted(regression_results[["CNPF.PS"]]),
    MER_Fitted = fitted(regression_results[["MER.PS"]]),
    ABS_Fitted = fitted(regression_results[["ABS.PS"]]),
    TEL_Fitted = fitted(regression_results[["TEL.PS"]])
  )

cat("FITTED VALUES ADDED\n")
```

FITTED VALUES ADDED

```
cat("New columns:", grep("Fitted", colnames(reg_data_wide), value = TRUE), "\n\n")
```

New columns: BPI_Fitted CNPF_Fitted MER_Fitted ABS_Fitted TEL_Fitted

4.2 Question 2 (5 pts)

Still in the same dataframe, compute the residuals for each stock.

$$\hat{u}_{i,t} = R_{i,t} - \hat{R}_{i,t}$$

```
reg_data_wide <- reg_data_wide |>
  mutate(
    BPI_Residual   = BPI.PS   - BPI_Fitted,
    CNPF_Residual  = CNPF.PS  - CNPF_Fitted,
    MER_Residual   = MER.PS   - MER_Fitted,
    ABS_Residual   = ABS.PS   - ABS_Fitted,
    TEL_Residual   = TEL.PS   - TEL_Fitted
  )

cat("RESIDUALS ADDED\n")
```

RESIDUALS ADDED

```
cat("New columns:", grep("Residual", colnames(reg_data_wide), value = TRUE), "\n\n")
```

New columns: BPI_Residual CNPF_Residual MER_Residual ABS_Residual TEL_Residual

4.3 Question 3 (3 pts)

What did you notice about the sum of the residuals for each stock?

```
residual_sums <- reg_data_wide |>
  summarise(
    BPI_Sum   = sum(BPI_Residual, na.rm = TRUE),
    CNPF_Sum  = sum(CNPF_Residual, na.rm = TRUE),
    MER_Sum   = sum(MER_Residual, na.rm = TRUE),
    ABS_Sum   = sum(ABS_Residual, na.rm = TRUE),
    TEL_Sum   = sum(TEL_Residual, na.rm = TRUE)
  )

cat("SUM OF RESIDUALS\n")
```

SUM OF RESIDUALS

```
knitr::kable(residual_sums,
              col.names = c("BPI", "CNPf", "MER", "ABS", "TEL"),
              digits = 6,
              booktabs = TRUE,
              caption = "Sum of Residuals by Stock")
```

Table 2: Sum of Residuals by Stock

BPI	CNPf	MER	ABS	TEL
0	0	0	0	0

```
cat("\n")
```

```
for (ric_name in c("BPI", "CNPf", "MER", "ABS", "TEL")) {
  col_name <- paste0(ric_name, "_Sum")
  sum_val <- residual_sums[[col_name]]
  cat(ric_name, ": ", sprintf("%.6f", sum_val), "\n", sep = "")
}
```

```
BPI: 0.000000
CNPf: -0.000000
MER: 0.000000
ABS: 0.000000
TEL: 0.000000
```

```
cat("\n")
```

The sum of residuals for every stock is close to zero as a result of their respective residuals cancelling out.

5 Data Export

Date, PSEi_Return, BPI.PS, MER.PS, TEL.PS, CNPf.PS, ABS.PS,
BPI_Fitted, CNPf_Fitted, MER_Fitted, ABS_Fitted, TEL_Fitted,
BPI_Residual, CNPf_Residual, MER_Residual, ABS_Residual, TEL_Residual

6 Summary of Submitted Files

#	File	Description
1	H3_Group4_Regression_Original.csv	Input data frame (727 rows $\times$ 17 cols)
2	H3_Analysis.R	Standalone R analysis script
3	H3.pdf	This document — PDF with answers